

Tutorial 3:

Question 1: Name three lossy compression methods

The three lossy compressions are JPEG, MPEG, and MP3.

Question 2: When would you use JPEG? When would you use MPEG?

JPEG is used to compress images while MPEG is used to compress video.

Question 3: Describe various stages of the JPEG algorithm.

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Question 4: Why is the discrete cosine transform needed in JPEG?

DCT changes the 64-pixel values in each block so that the relative relationships between pixels are kept but the redundancies are removed.

Question 5: Why we change RGB space to YCbCr

Medical investigation of the eye has led to findings that the rods some 120 million in number, are much more sensitive than the cones which are around 6-7 million in number. The rods are not sensitive to colour, while the cones which provide much of the eye's colour sensitivity are found to be located close to a central region

Question 6: In JPEG, what is the function of blocking?

Blocking is the act of dividing the image into 8×8 blocks in order to reduce the number of calculations.

Question 7: How does quantisation contribute to compression?

Quantization of the T table reduces the number of bits needed for encoding each value.